

Part B – Data Analysis & Testing Tasks

1. Hypothesis Formulation

Hypothesis 1

H_0 :Smoking has no effect on Diabetes

H_1 :Smoking affects Diabetes

Hypothesis 2

H_0 :Mean BMI is same across all age groups

H_1 :Mean BMI differs across age groups

2. Confidence Interval (Example: Age)

Given:

Sample Mean

$$\bar{X} = 40$$

Standard Deviation

$$\sigma = 10$$

Sample Size

$$n = 200$$

Confidence Level = 95%

Critical Value

$$Z = 1.96$$

Step 1 – Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$
$$SE = \frac{10}{\sqrt{200}}$$
$$SE = \frac{10}{14.14}$$
$$SE = 0.707$$

Step 2 – Margin of Error

$$ME = Z \times SE$$
$$ME = 1.96 \times 0.707$$
$$ME = 1.38$$

Step 3 – Confidence Interval

$$CI = \bar{X} \pm ME$$

Lower Limit

$$40 - 1.38 = 38.62$$

Upper Limit

$$40 + 1.38 = 41.38$$

Final CI

$$(38.62, 41.38)$$

3. Critical Value

Significance Level

$$\alpha = 0.05$$

Two-tailed test

$$Z_{\frac{\alpha}{2}} = 1.96$$

Decision Rule

$$|Z| > 1.96 \rightarrow \text{Reject } H_0$$

4. Z-Test (Example)

Given:

Sample Mean

$$\bar{X} = 110$$

Population Mean

$$\mu = 100$$

Standard Deviation

$$\sigma = 15$$

Sample Size

$$n = 200$$

Step 1 – Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$
$$SE = \frac{15}{\sqrt{200}}$$

$$SE = 1.06$$

Step 2 – Z Statistic

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{110 - 100}{1.06}$$

$$Z = \frac{10}{1.06}$$

$$Z = 9.43$$

Decision

$$9.43 > 1.96$$

Reject H_0

5. Chi-Square Test

Observed Table

Smoking	Diabetes	No Diabetes	Total
Smoker	40	60	100
Non-Smoker	20	80	100

Total = 200

Step 1 – Expected Frequency

Formula

E

$$= \frac{\text{Row } T \times \text{Column } T}{\text{Grand } T}$$

Example

$$E = \frac{100 \times 60}{200}$$

$$E = 30$$

Step 2 – Chi-Square Formula

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Calculation

$$\frac{(40-30)^2}{30} = 3.33$$

$$\frac{(60-70)^2}{70} = 1.43$$

$$\frac{(20-30)^2}{30} = 3.33$$

$$\frac{(80-70)^2}{70} = 1.43$$

Total Chi-Square

$$\chi^2 = 9.52$$

6. ANOVA Test**Step 1 – Mean of Groups**

$$\bar{X}_1,$$

$$\bar{X}_2,$$

$$\bar{X}_3$$

Step 2 – Sum of Squares Between

$$SS_{between} = \sum n_i (\bar{X}_i - \bar{X})^2$$

Step 3 – Sum of Squares Within

$$SS_{within} = \sum (X_{ij} - \bar{X}_i)^2$$

Step 4 – Mean Squares

$$MS_{between} = \frac{SS_{between}}{df_{between}}$$

$$MS_{within} = \frac{SS_{within}}{df_{within}}$$

Step 5 – F Statistic

$$F = \frac{MS_{between}}{MS_{within}}$$

7. Covariance

Formula

$$Cov(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Example Data

Age

20, 25, 30

BMI

22, 24, 27

Step 1 – Mean

$$\bar{x} = 25$$

$$\bar{y} = 24.33$$

Step 2 – Calculation

$$(20 - 25)(22 - 24.33)$$

$$(25 - 25)(24 - 24.33)$$

$$(30 - 25)(27 - 24.33)$$

$$Cov = \frac{25}{2}$$

$$Cov = 12.5$$

8. Correlation

Formula

$$r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Example

$$Cov = 12.5$$

$$\sigma_X = 5$$

$$\sigma_Y = 3$$

Calculation

$$r = \frac{12.5}{5 \times 3}$$

$$r = \frac{12.5}{15}$$

$$r = 0.83$$